23/12/24 M/S

USN

[23CSE309]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2024

INTERNET OF THINGS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Define IOT.	L1	CO1	**02**
	b.	Explain the Arduino IDE.	L1	CO2	**02**
	c.	Recall the MAC address in the context of IoT communication.	L1	CO3	**02**
	d.	List the operational view specifications in IOT.	L1	CO4	**02**
	e.	Name two advantages of using MapReduce for IoT data analysis.	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain machine to machine communications (M2M) in detail.	L2	CO1	**05**
		OR			
	b.	Discuss about different IOT levels with neat sketch.	L3	CO1	**05**
3.	**a.**	Explain IOT devices with examples.	L2	CO2	**05**
		OR			
	b.	Give the difference between Arduino and Raspberry Pi.	L2	CO2	**05**
4.	**a.**	Define the following terms with respect to IoT: **i.** IP addresses **ii.** DNS **iii.** Static IP address assignment **iv.** Dynamic IP address assignment	L2	CO3	**05**
		OR			
	b.	Difference between active and passive sensors and explain the sonic sensor working details.	L2	CO3	**05**
5.	**a.**	Explain the component and device integration.	L2	CO4	**05**
		OR			
	b.	What is a smart home? What are the functions of IoT based smart home? Which sensors are required to make smart home?	L2	CO4	**05**
6.	**a.**	Describe YARN Components in detail.	L3	CO5	**05**
		OR			
	b.	What is Apache Oozie? Explain its work flow in detail.	L2	CO5	**05**
7.	**a.**	Explain the Web socket based API's in IOT.	L2	CO1	**05**
		OR			
	b.	Discuss about hadoop schedulers.	L3	CO5	**05**

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		PART-C			6 X 10=60
8.	**a.**	What are the four pillars of IOT? Describe each pillar with the help of suitable example.	L2	CO1	**10**
		OR			
	b.	Discuss about various protocols in IOT.	L2	CO1	**10**
9.	**a.**	Explain Raspberry Pi Model B. What are the things need to be considered for developing on the Raspberry Pi?	L3	CO2	**10**
		OR			
	b.	Explain in detail about various Domain Specific IoT applications.	L2	CO2	**10**
10.	**a.**	Discuss about Internet communications with neat sketch.	L3	CO3	**10**
		OR			
	b.	Explain the role of Sensors and its categories in IOT applications.	L2	CO3	**10**
11.	**a.**	Illustrate the design of IoT for smart cities environment with automations.	L3	CO4	**10**
		OR			
	b.	Describe the necessary steps required for design methodology in IOT.	L2	CO4	**10**
12.	**a.**	Explain the following **i.** How does Oozie work. **ii.** Oozie work flow diagram example. **iii.** Packing and deploying an Oozie workflow application. Features of Oozie.	L2	CO5	**10**
		OR			
	b.	Explain the Hadoop Ecosystem with neat sketch.	L2	CO5	**10**
13.	**a.**	With neat sketch, describe the structure of IOT device.	L2	CO1	**10**
		OR			
	b.	Discuss about process specification in IOT.	L2	CO4	**10**